

Lab Reference: BIOS and UEFI Differences

System Administration and Maintenance

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1. Breakdown Table

Feature	Legacy BIOS	UEFI Firmware
Processor Mode	16-bit Real Mode	32-bit or 64-bit Mode
Partition Style	MBR (Master Boot Record)	GPT (GUID Partition Table)
Max Storage Size	Up to 2.2 TB only	Up to 9.4 ZB
Partition Count	Max 4 Primary partitions	Up to 128 Primary partitions
Boot Security	Basic Password setup only	Secure Boot (Digital Signatures)
User Interface	Blue/Grey Text UI (Keyboard only)	Modern GUI (Mouse/Touch support)
Boot Storage Path	First sector of the drive (Sector 0)	EFI System Partition (FAT32 format)

2. Practical Notes & Observations

Legacy BIOS Overview

- Runs a basic Power-On Self-Test (POST) when turning on the PC to check if RAM and core hardware respond.

- Stuck with 1 MB memory addressing, which limits how fast the motherboard can load boot code.
- Reads directly from the MBR sector, making it prone to bootloader overwrite issues if installed incorrectly.

UEFI Overview

- Loads network drivers before the OS boots, allowing remote diagnostics and PXE boot setups.
- Uses Secure Boot to block unauthorized bootloaders or malicious code from executing early in the sequence.
- Keeps executable boot files (like .efi scripts) inside an isolated FAT32 partition instead of raw disk sectors.